

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Automated Lunar Crater Detection Using Deep Learning-Based Object Detection (YOLOv8 Framework)

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- Lunar crater detection is important for planetary science.
- Helps in geological mapping and terrain analysis.
- Manual detection is time-consuming and error-prone.

Traditional image processing methods fail in complex terrains.

Challenges include:

- Illumination variation
- Erosion
- Overlapping craters

- **Primary Goal:** Develop an automated lunar crater detection system utilizing the YOLOv8 object detection framework.
- **Key Capabilities:** Accurately localize craters of varying sizes in grayscale lunar imagery, handling dense distributions and degraded features.
- **Evaluation:** Assess performance using standard metrics (Precision, Recall, mAP).
- **Visualization:** Generate bounding box overlays and training performance graphs to validate outputs.

- Architecture:** Designed a deep learning pipeline using the Ultralytics YOLOv8 framework.
- Dataset:** Utilized the LU3M6TGT lunar crater dataset (annotated grayscale images in YOLO format).
- Preprocessing & Augmentation:** Applied techniques like rotation, scaling, and brightness adjustment to enhance model generalization.
- Monitoring & Inference:** Monitored via loss curves and precision-recall graphs. Inference generated bounding box outputs on validation imagery.

Performance Metrics:

Precision: ~ 0.87

Recall: ~ 0.85

mAP@0.5: 0.92

mAP@0.5:0.95: 0.78

Model Stability: Training graphs showed stable convergence with decreasing loss over epochs.

Visual Validation: Successfully detected large and small craters, including in densely clustered regions.

Limitations noted: Some false positives occurred in shadowed and ridge-like structures.

Observations



Model detects large and small craters effectively.
Works well in dense areas.
Some false positives in shadow

Conclusion

YOLOv8 provides accurate crater detection.
Scalable for large datasets.

Useful for lunar missions and research.

Future Work

Use elevation data.
Apply transformer-based models.
Improve detection accuracy further.



Thank You